Lecture 1: Introduction

Syllabus

- **Location and Time**

  TT403 on Thursday (9:00-9:50 am)

  TT404 on Friday (9:00-10:50 am)

- **Prerequisites**

  ENGG1300 Fundamental Mechanics

- **Consultation Hours (Prof. H. Ye)**

  Wednesday, 11:00 am- 12:00 pm (by appointment)

  Additional consultation:

  before homework submission and the final exam revision week

Syllabus

- **Grading**

  Final examination (60%)

  Continuous assessment (Assignment) (30%)

  Practical work (Laboratory work) (10%)

  Absence from Laboratory Session or Test:

  Absence from laboratory session or test with no valid reason will result in failure of this course

  Late Submission of Assignment:

  Deduction in mark according to the latest departmental policy

Teaching Schedule

Lecturers:

- Prof. H. Ye (hlye@hku.hk)
- Prof. J. Wang (cewang@hku.hk)

Teaching Assistant of Prof. H. Ye:

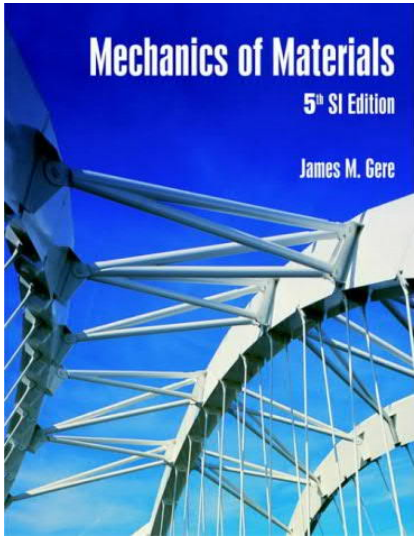
- Haojia ZHONG (zhonghj@connect.hku.hk)
- Yuquan HU (yuquanhuce@connect.hku.hk)

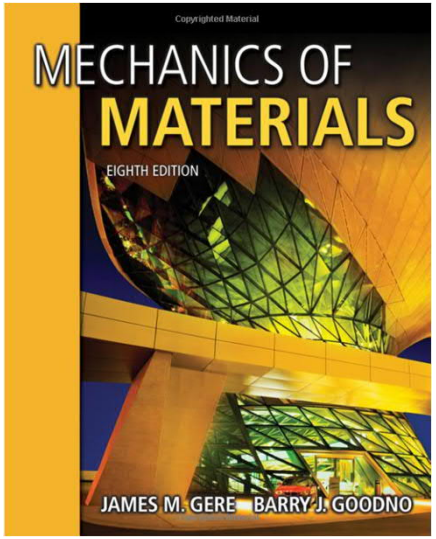
Thursday (9:00 - 9:50am) Venue: TT403		Friday (9:00 - 10:50am) Venue: TT404	
Sep 3	Prof Ye Introduction	Sep 4	Prof Ye Axial loading
Sep 10	Prof Ye Axial loading	Sep 11	Prof Ye Internal forces and moment
Sep 17	Prof Ye Internal forces and moment	Sep 18	Prof Ye Beams in Symmetric Bending
Sep 24	Prof Ye Beams in Symmetric Bending	Sep 25	Prof Wang Normal stress of unsymmetric section
Oct 1	**Holiday**	Oct 2	Prof Wang Normal stress of composite section
Oct 8	Prof Wang Shear stress	Oct 9	Prof Wang Shear stress
Oct 15	**Reading Week**	Oct 16	**Reading Week**
Oct 22	Prof Wang Torsion	Oct 23	Prof Wang Torsion
Oct 29	Prof Wang Stress transformation	Oct 30	Prof Wang Stress transformation
Nov 5	Prof Wang Stress transformation	Nov 6	Prof Wang Flexural deformation
Nov 12	Prof Wang Flexural deformation	Nov 13	Prof Wang Flexural deformation
Nov 19	Prof Wang Failure criterions	Nov 20	Prof Wang Failure criterions
Nov 26	Prof Wang Column buckling	Nov 27	Prof Wang Column buckling

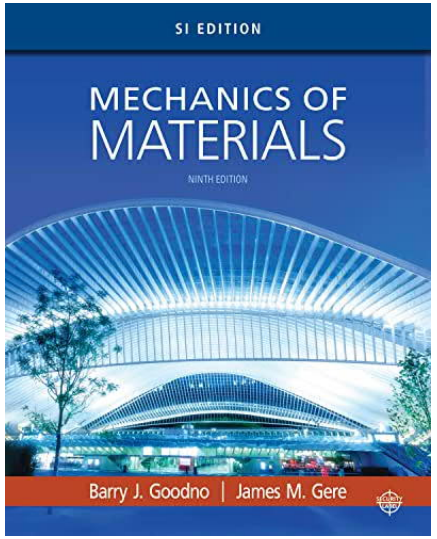
Reference

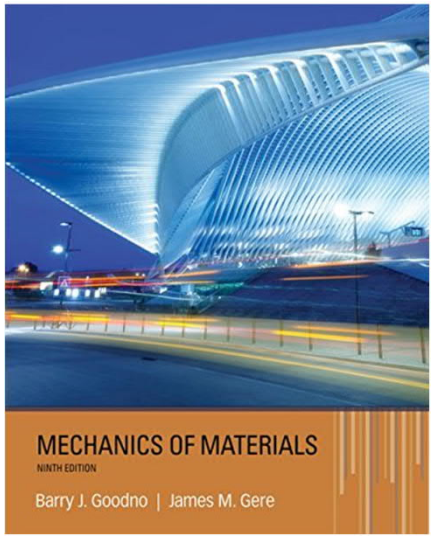
1. **Gere, J., & Goodno, B. J. (2012). *Mechanics of materials.* Nelson Education. [5th edition to 9th edition are all fine, recommended]**
2. Beer, F. P., Johnston, E. R., DeWolf, J. T., & Mazurek, D. F. (2009). *Mechanics of materials*, 5th edn New York. NY: McGraw-Hill.
3. Hibbeler, R. C. (2013). *Statics and mechanics of materials.* Pearson Higher Ed.

Note: also named as **Strength of Materials**









Course Description

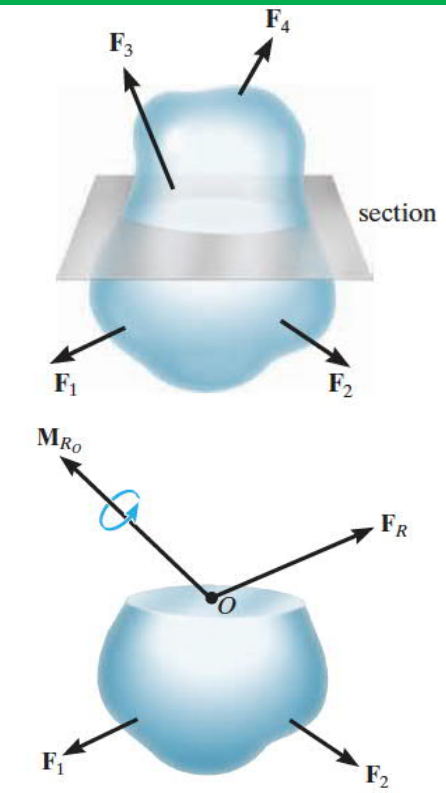
Analyze behaviors of deformable bodies subject to external forces

Objective

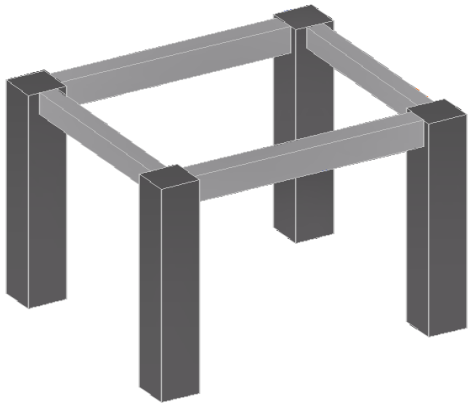
To study strength of materials is to provide the future engineer with the means of analyzing and designing various machine and load bearing structures.

Course Description

- External forces applied to a deformable body that induces stress, deformation, and strain;
- Development of basic relationship between load, stress, strain and deformation of structure under tension, compression, shear, torsion, bending, and their combination;
- Failure criterion: **Strength** (Load-carrying capacity), **Stiffness** (resistance to deformation) and **Stability** (ability to maintain configuration) of structure (3S);



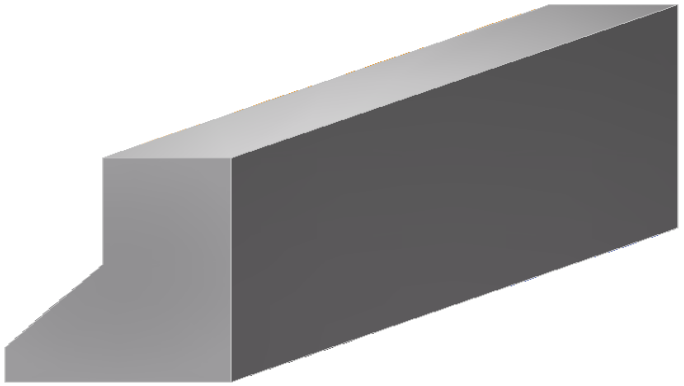
Classification of Structures



Bar (Beam, Column, Frame, Cable): one dimension is far greater than the others



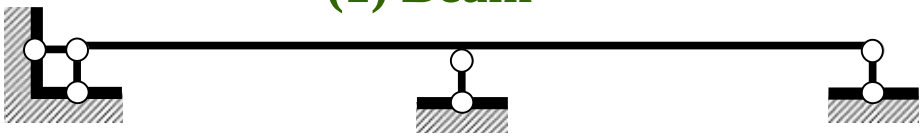
Plate (Wall, Floor, Membrane): two dimensions are far greater than the third.



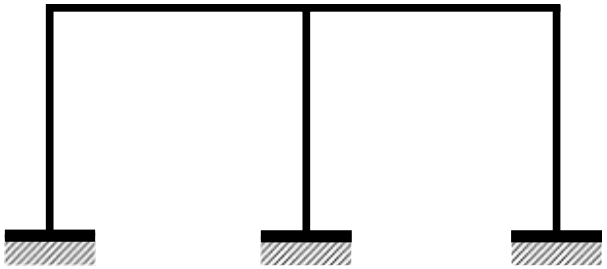
Solid: three dimensions are equivalent

Classification of Bar Structures

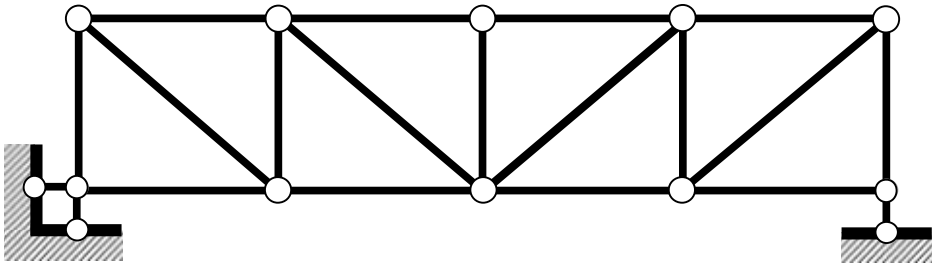
(1) Beam



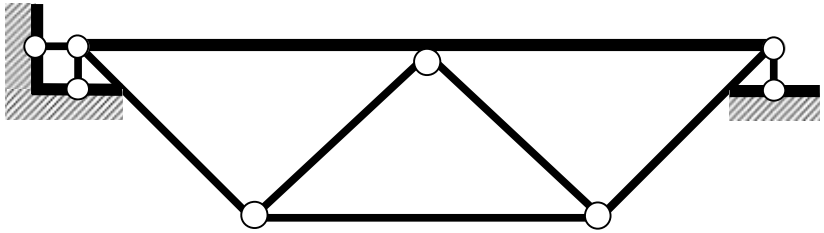
(2) Frame



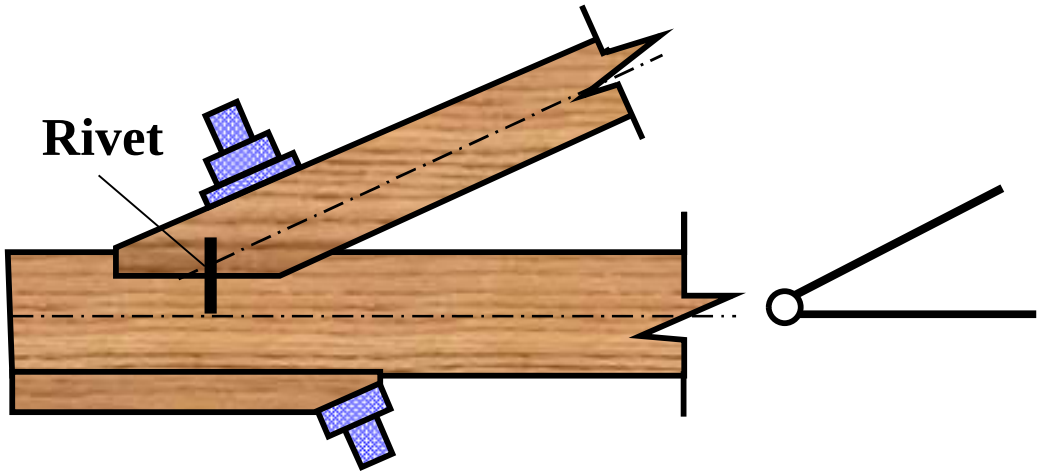
(3) Truss



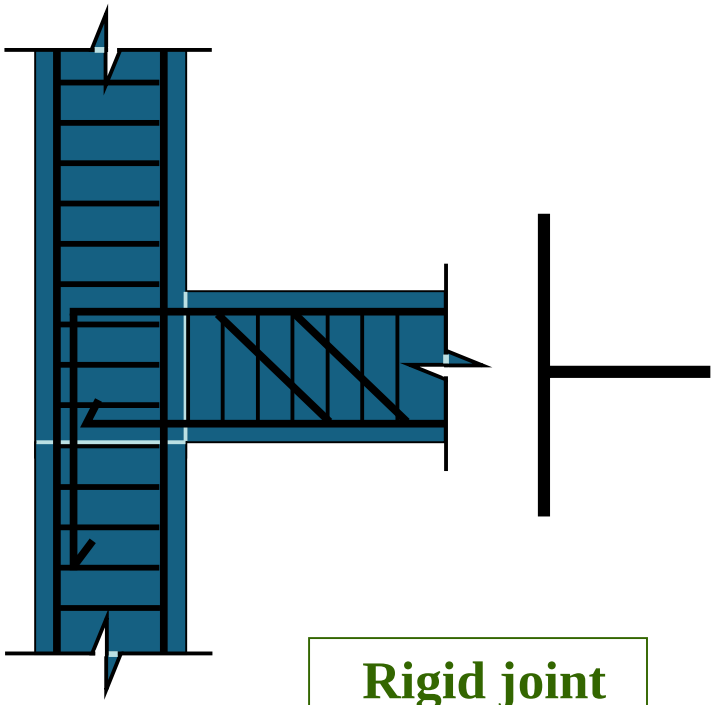
(4) Compound Structure



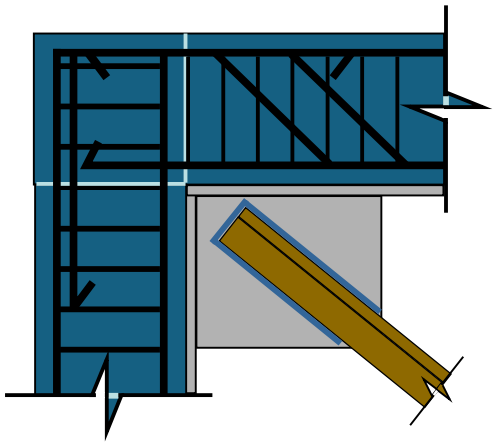
Types of Joint Connections

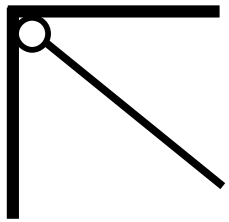


Pinned joint



Rigid joint





Composite Joint

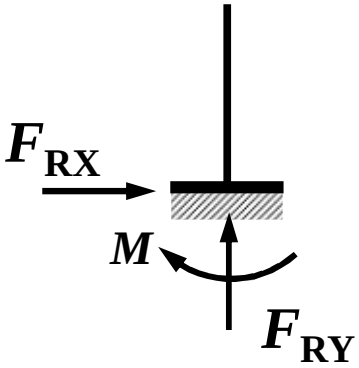
Types of Supports



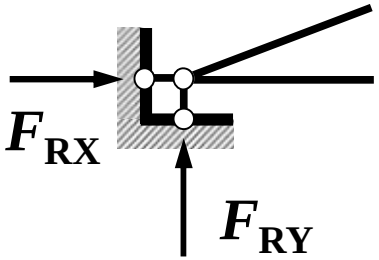


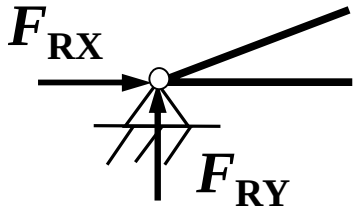


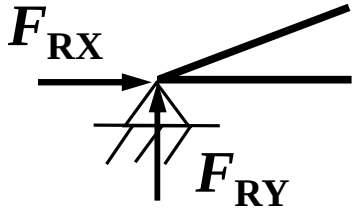
Symbols do not represent the actual physical construction



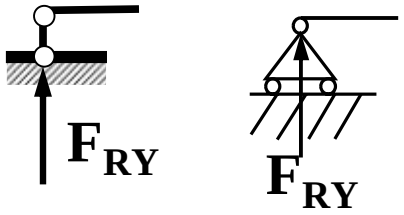
Fixed Support

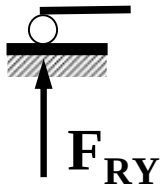






Pinned support

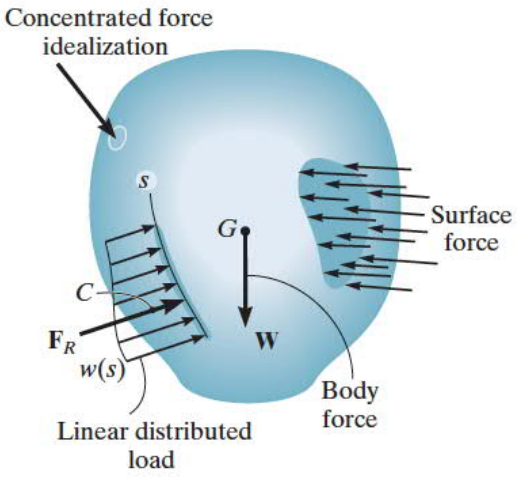




Roller support

Types of Loads

Criteria	Classification
Exerting Object	Active Load/Reaction
Size of Acting Scope	Concentrated/ Distributed Load
Dimensions of Distributed Loads	Volume/Surface/ Linear Loads
Acting Period	Instantaneous (Live)/ Constant (Dead) Load
Time Dependency	Static/Dynamic Load
Characteristics	External Load/ Internal Force

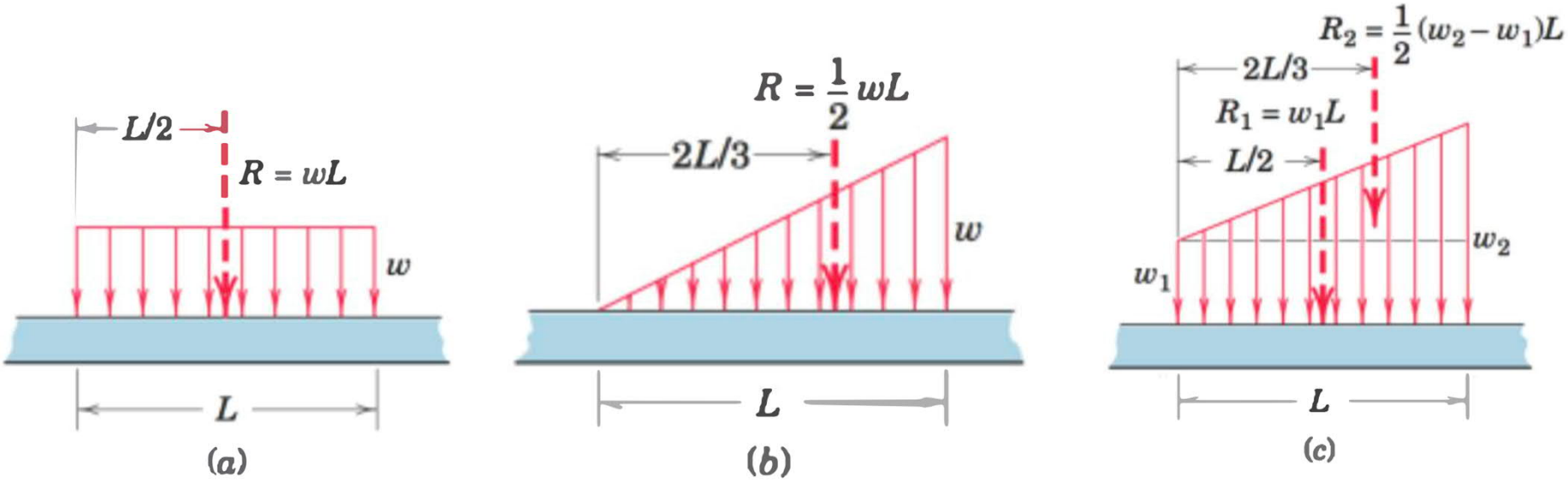


Types of Loads

Concentrated load: a load is applied over a very small area (a point)
Distributed load: a load is spread along the axis of a beam

- Distributed loads are measured by their intensity, which is expressed in units of [Force/Distance] (for example, newtons per meter)

Concentrated moment: force couple to create rotation without translation

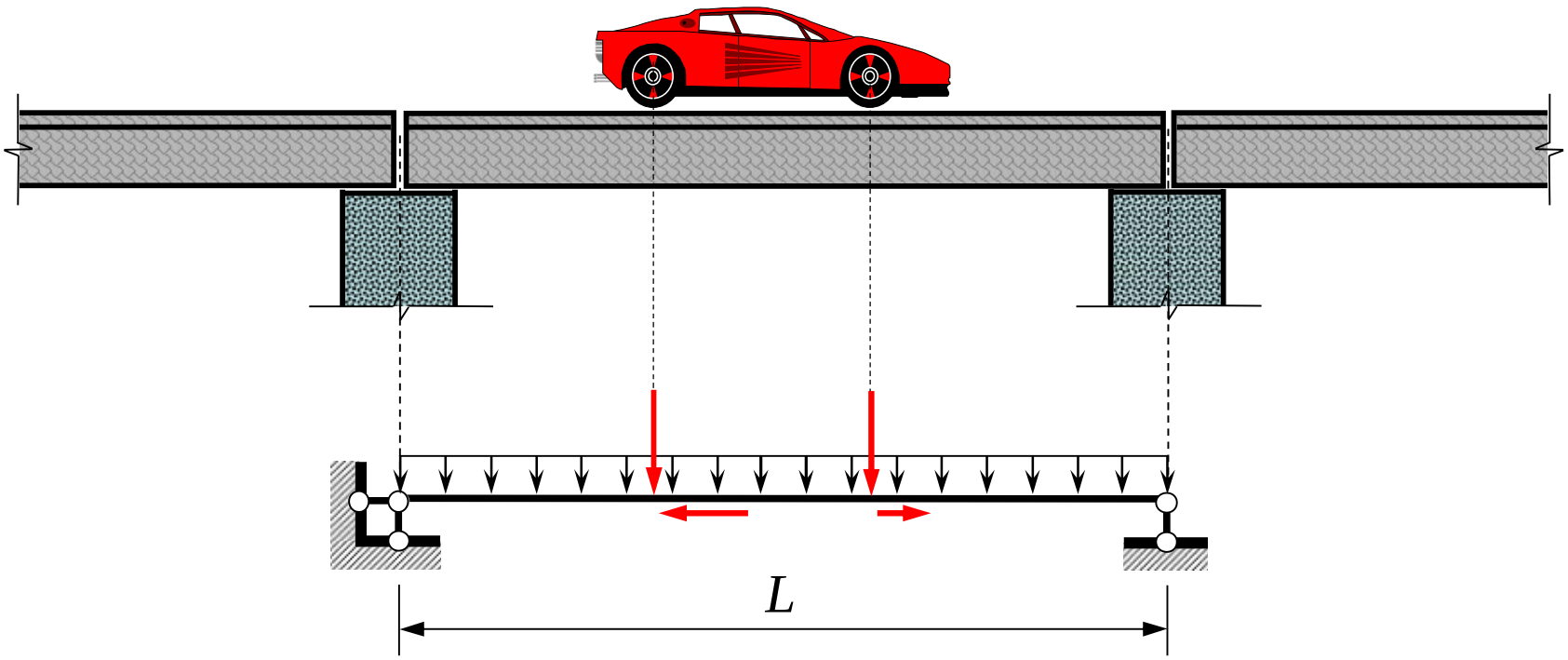


Uniformly distributed

Linearly varying distributed

- The **equivalent concentrated load** is a single concentrated force that will have **the same effect** on a body as the original distributed loads

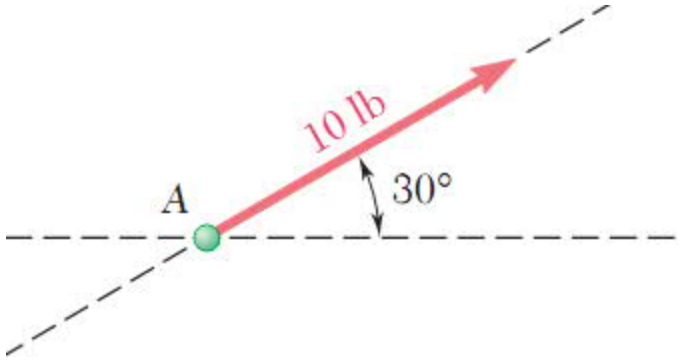
Analytical Model: Simplification of Physical Model



Simple Supported Beam (for rear wheel drive)

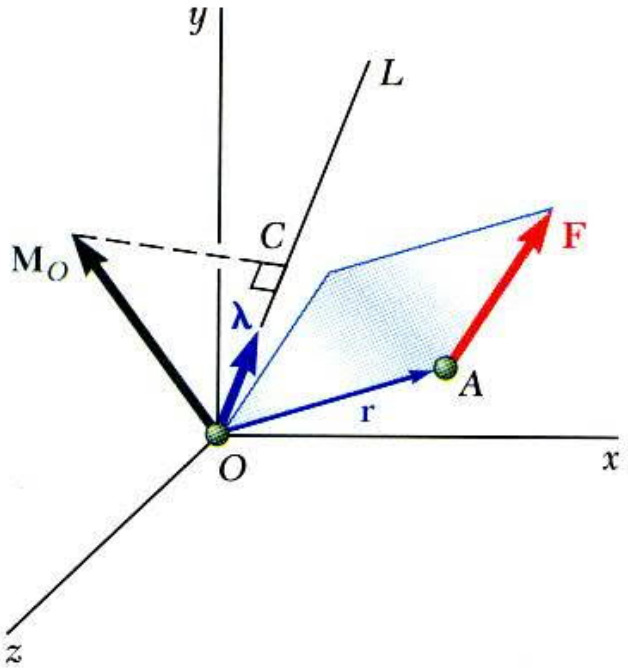
Force and Moment

- **Force:** action ***F*** of one body on another; represented by line of action and characterized by its point of application, magnitude, and direction.



- **Moment:** $\boldsymbol{M_O}$ is a force ***F*** applied at the point *A* about a point *O*,

$$M_O = r \times F$$

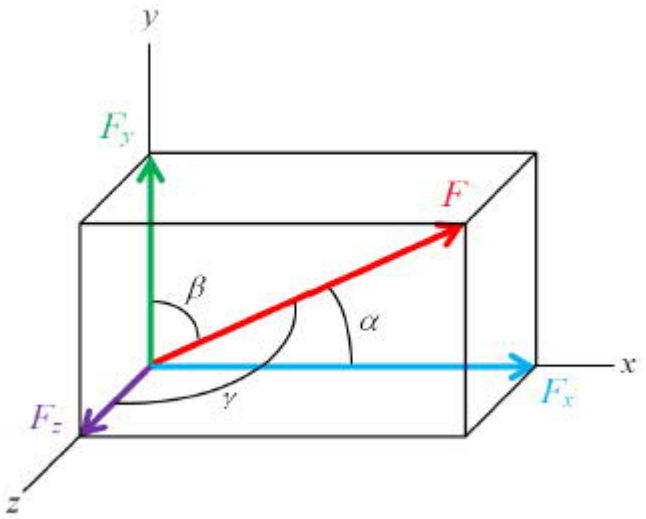


Decomposition of Force and Moment

- **Force Decomposition:**

  Force ***F*** can be decomposed into ***x***, ***y*** and ***z*** components as

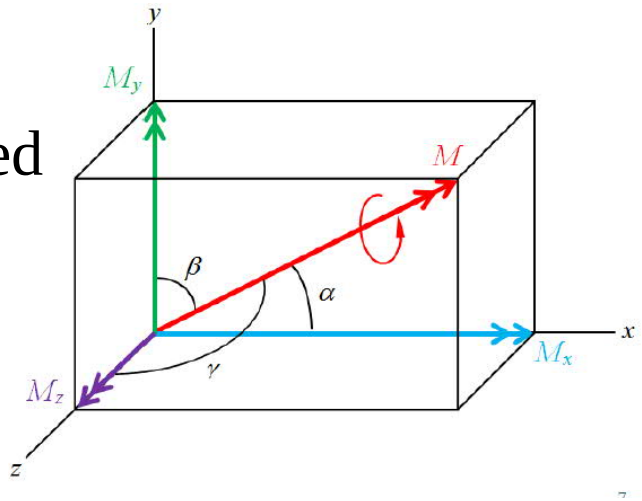
$$F_x = F\cos\alpha; \quad F_y = F\cos\beta; \quad F_z = F\cos\gamma$$



- **Moment Decomposition:**

  Moment ***M*** about point O can be decomposed into ***x***, ***y*** and ***z*** components as

$$M_x = M_o\cos\alpha; \quad M_y = M_o\cos\beta; \quad M_z = M_o\cos\gamma$$



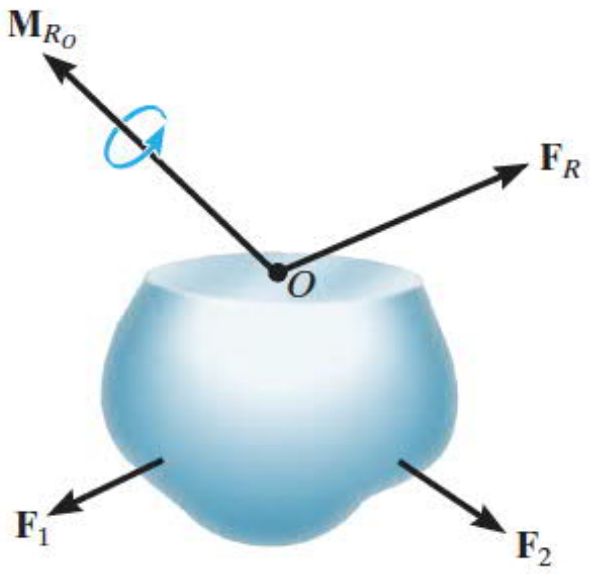
Equilibrium

Equilibrium of Structures (3D)

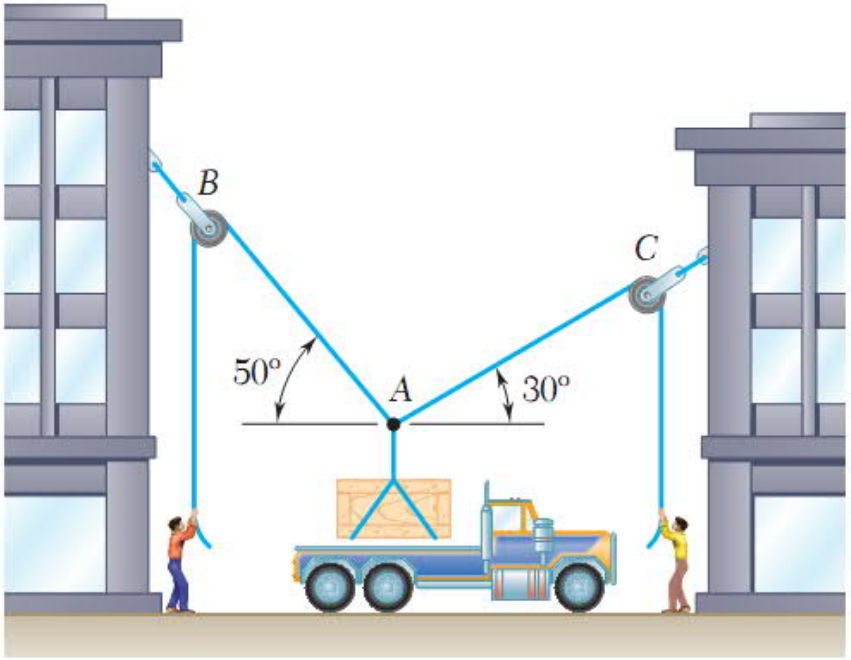
- A structure is considered to be in Equilibrium, if, initially at rest, it remains at rest when subjected to a system of forces and moments (couples);
- Equilibrium conditions:

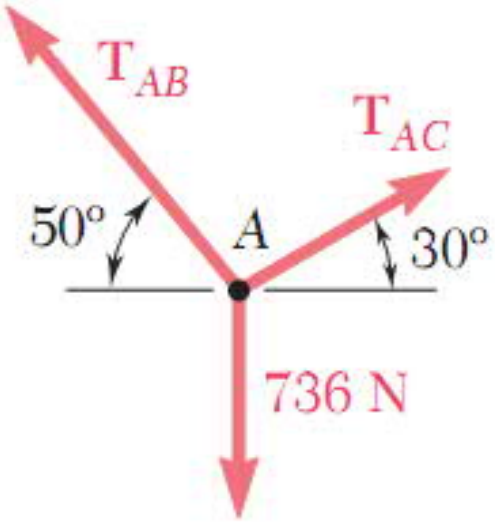
$$\sum F_x = 0; \quad \sum F_y = 0; \quad \sum F_z = 0$$

$$\sum M_x = 0; \quad \sum M_y = 0; \quad \sum M_z = 0$$



Space Diagram vs. Free-Body Diagram





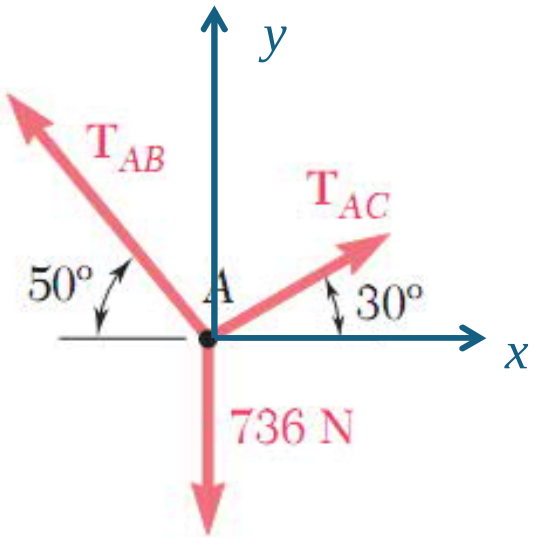
Space Diagram:

A sketch showing the physical conditions of the problem.

Free-Body Diagram:

A sketch showing only the forces on the simplified body/structure.

Free-Body Diagram

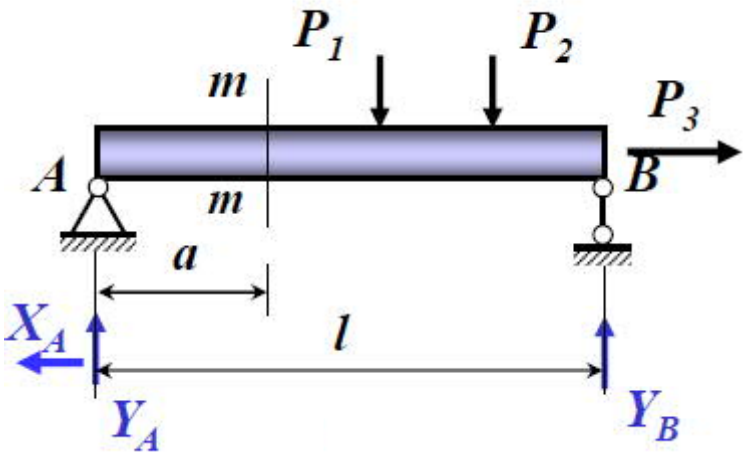


$$\begin{cases} \sum F_x = T_{AC}\cos(30^\circ) - T_{AB}\cos(50^\circ) = 0 \\ \sum F_y = T_{AC}\sin(30^\circ) + T_{AB}\sin(50^\circ) - 736 = 0 \end{cases}$$

External Force

External Force

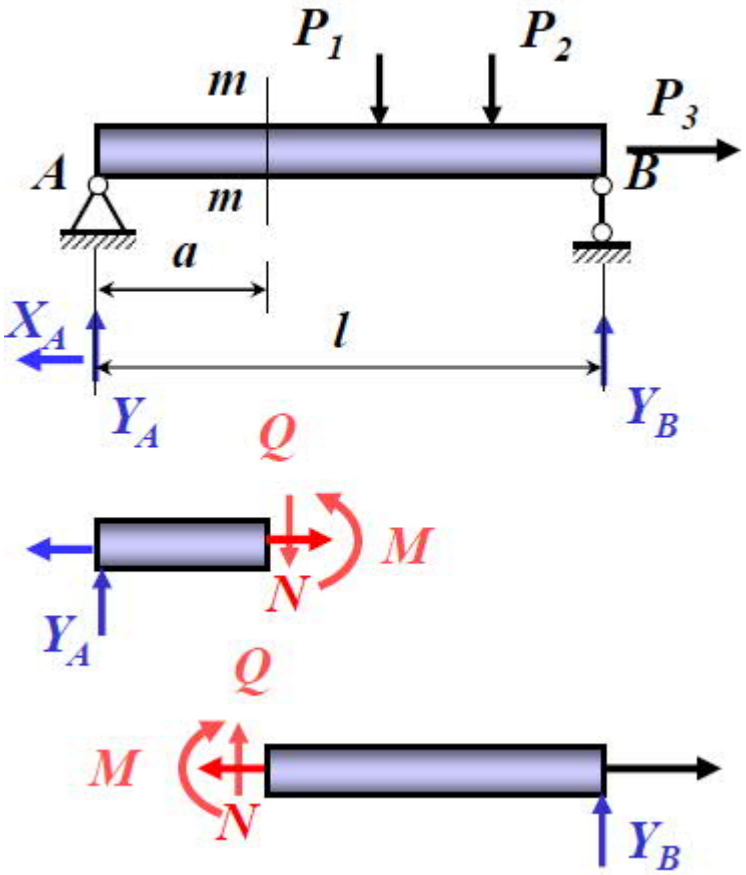
- Defined as the action of other bodies to structures;
- Classified as applied force (load) and support reaction.



Internal Force (Stress Resultants)

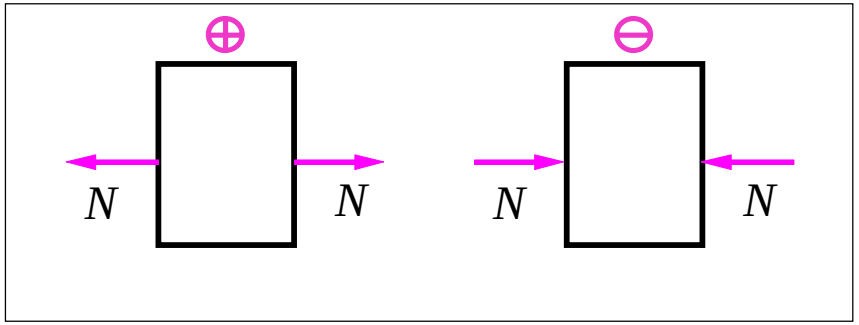
Internal Force

- Defined as the action to a portion of structure by the rest of structures;
- Internal forces vary along the bar and can be determined by **method of sections;**
- Internal forces are **the forces on the cross-sections** of bar structures;
- Internal forces include axial force, shear force and bending moment (for 2D case).

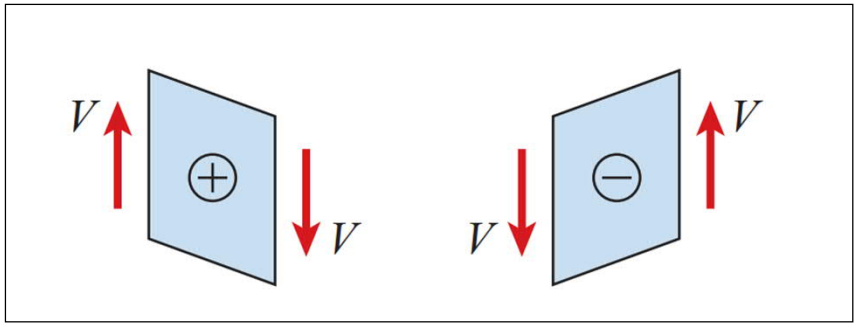


Sign Convention of Internal Force

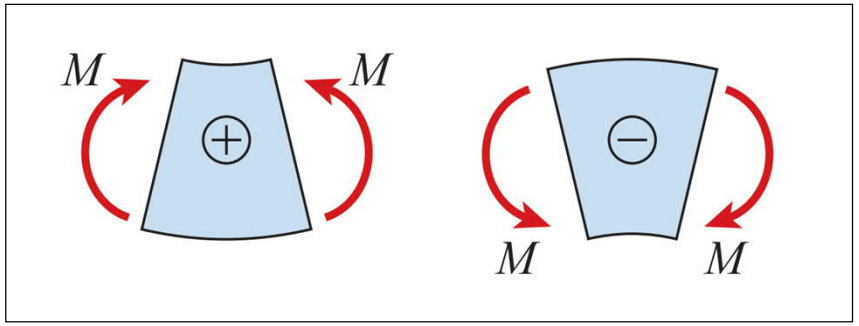
- **Axial force**: positive for tension and negative for compression;



- **Shear force**: positive for clockwise and negative for counterclockwise;

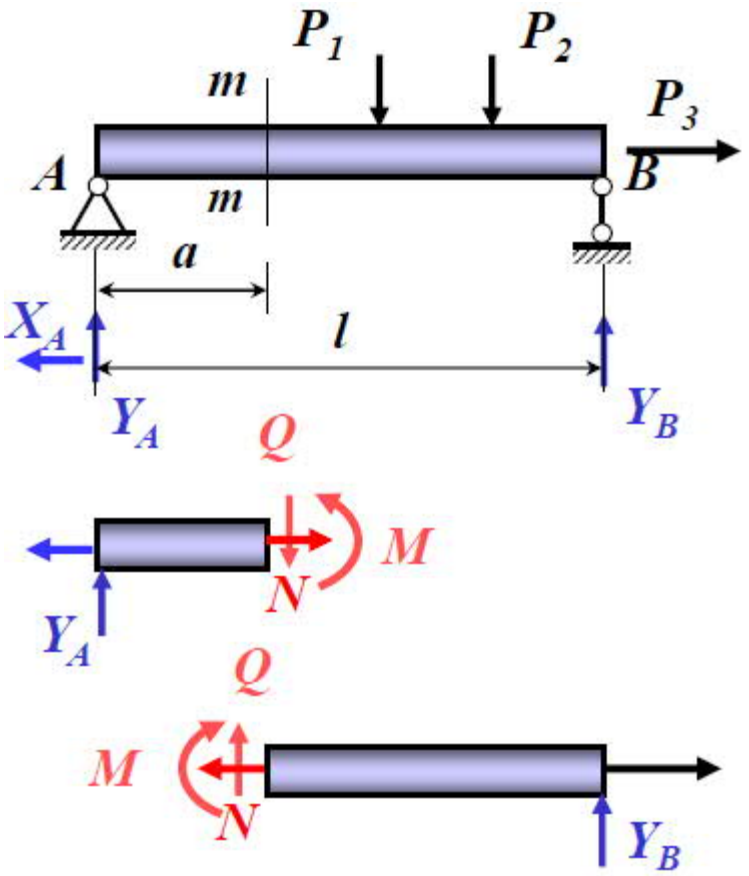


- **Bending Moment**: Positive: upper half under compression / lower half under tension (concave)



Sign Convention of Internal Force

- **Axial force**: positive for tension and negative for compression;
- **Shear force**: positive for clockwise and negative for counterclockwise;
- **Bending Moment**: Positive: upper half under compression / lower half under tension (concave)



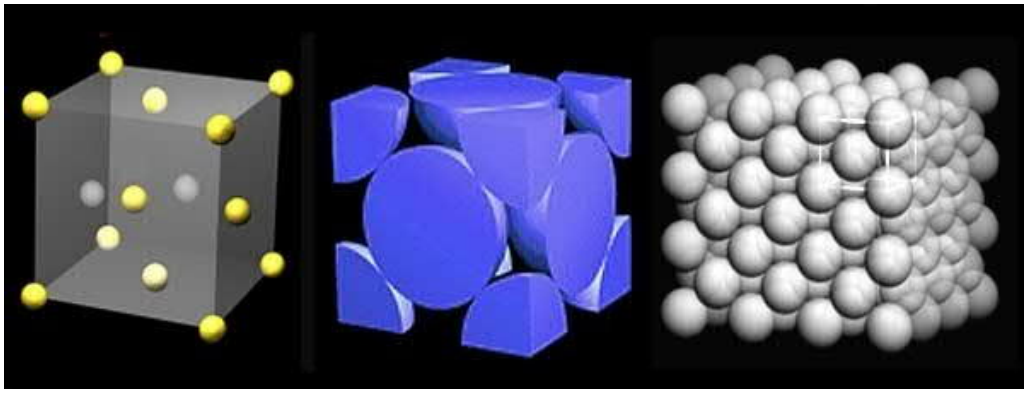
Note: figures on the right hand side shows the **positive** signs of internal forces.

Assumptions of Strength of Materials

- Continuity
- Homogeneity
- Isotropy
- Small Deformation
- Linear Elasticity

Continuity

- Matter fills up the whole space of a solid defined by its volume.
- Neither vacancies can be produced nor more materials can be added under normal working conditions.
- Arbitrary section or volume element can be extracted for force or deformation analysis.



Face-Centered Cubic Lattice

$$4r = a\sqrt{2} \Rightarrow r = \frac{\sqrt{2}}{4}a$$

$$V_{matter} = \frac{16}{3}\pi r^3 = \frac{16}{3}\pi\frac{2\sqrt{2}}{64}a^3 = \frac{\pi}{3\sqrt{2}}a^3$$

$$\Rightarrow \frac{V_{matter}}{V} = \frac{\pi}{3\sqrt{2}} \approx 74\%$$

Homogeneity vs Isotropy

Homogeneity

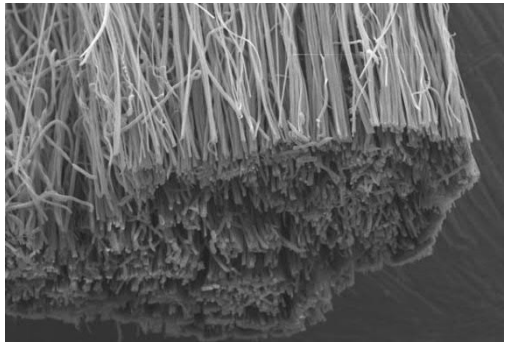
- Ideally, homogeneous material has the same properties at every point; it is uniform without irregularities.
- In reality, macroscopic material properties can be represented by any arbitrary *representative volume element* (RVE).



Cross section of wood

Isotropy

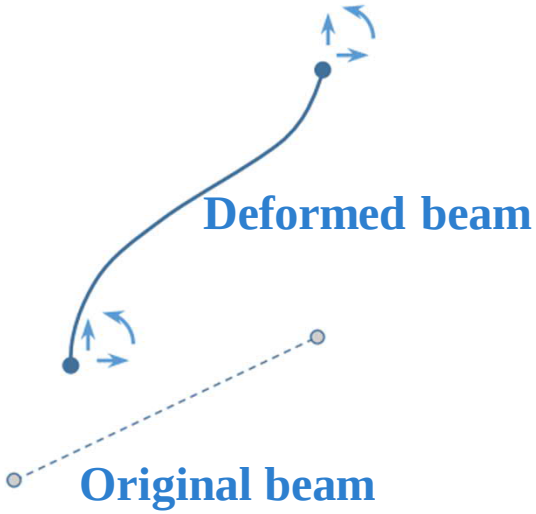
- Mechanical properties of materials are independent of directions.
- Strong transversely isotropic materials such as wood and fiber reinforced composites are still viewed as non-isotropic materials.



Fiber reinforced Composite

Small Deformation vs Large Deformation

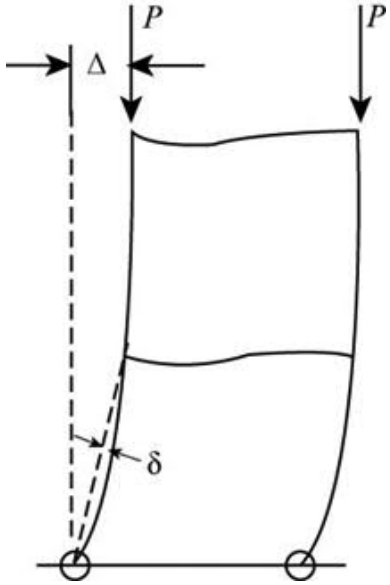
- **Small Deformation:** deformation of structural elements under mechanical loads are very small compared to their original size. Under small deformation, analysis of force and deformation can be based on a structure's **original size and shape** prior to deformation.
- **Large Deformation:** deform too large and request special attention. Tacoma bridge collapse is large deformation. **Second order effect**: due to lateral displacement, vertical load induce additional moment.



Large deformation



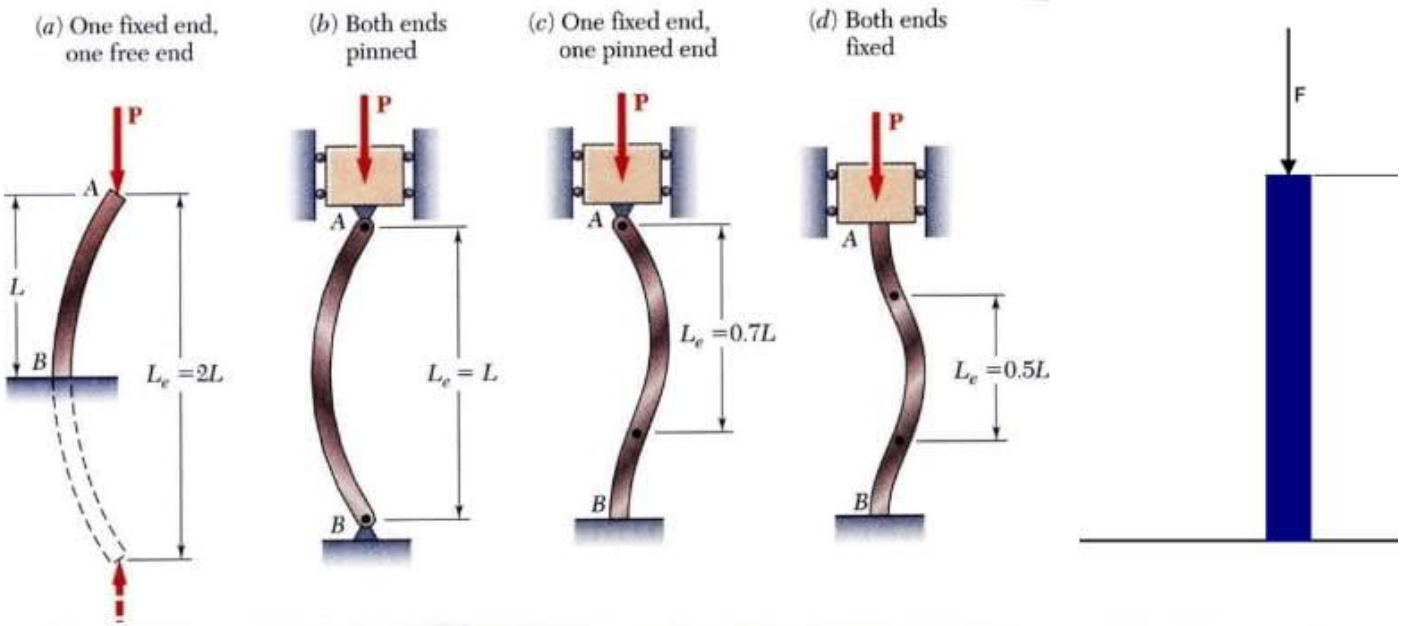
Wind-induced collapse (flutter instability)



Second order effect

Small Deformation vs Large Deformation

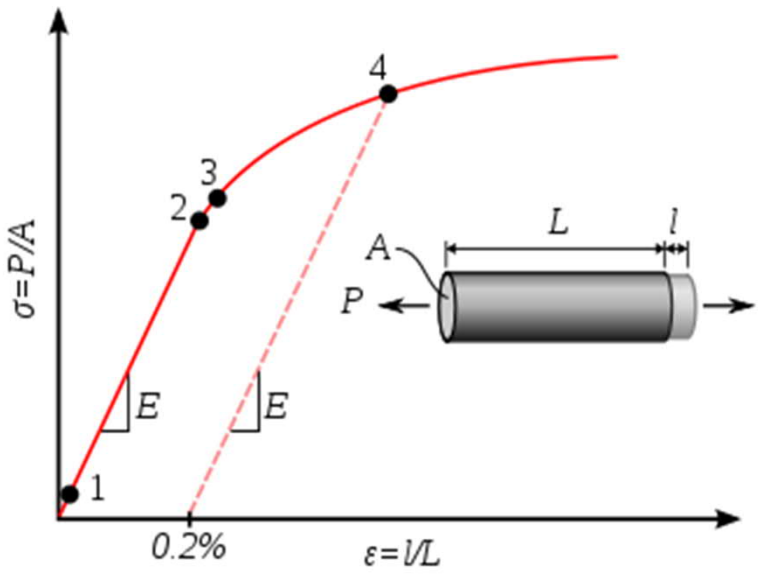
- **Large Deformation:** Buckling is a special form of large deformation.





Linear Elasticity

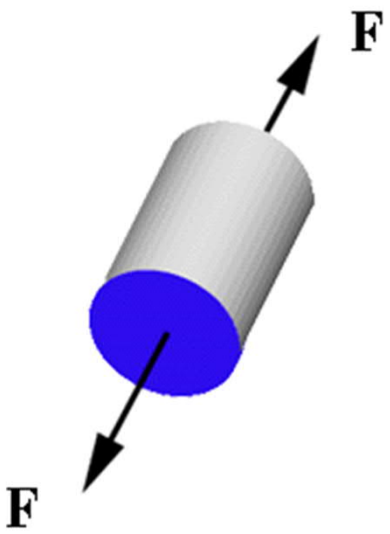
- **Elasticity:** a structural element can restore to its original size and shape upon the removing of its external loading.
- **Linear Elasticity:** deformation is *linearly proportional* to load.
- A linear elastic material obeys Hooke's law.



Stress-strain curve showing yield behavior.

1. A point within proportionality.
2. Proportionality limit.
3. Elastic limit (initial yield strength).
4. Subsequent yield strength.

Fundamental Actions of External Force





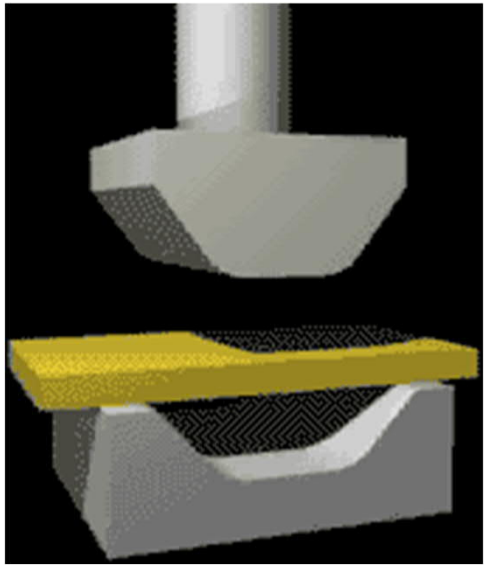
Tension (positive)

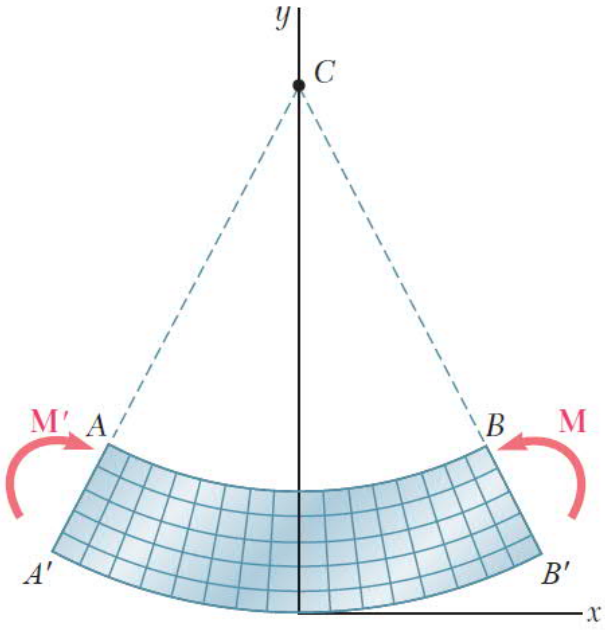
- Load: equal and opposite forces along bar axis
- Deformation: elongation along axis

Compression (negative)

- Load: equal and opposite forces along bar axis
- Deformation: contraction along axis

Fundamental Action of External Forces

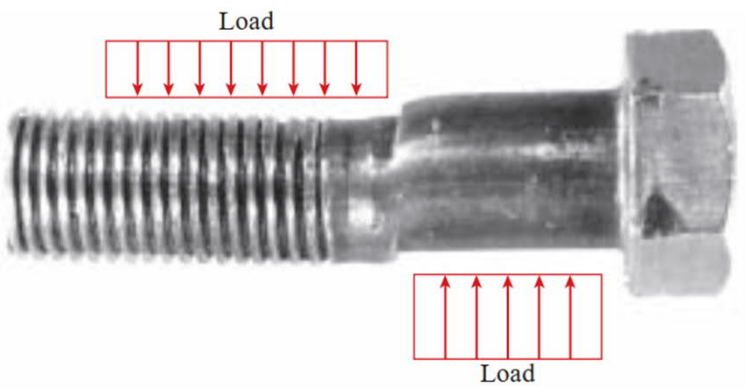


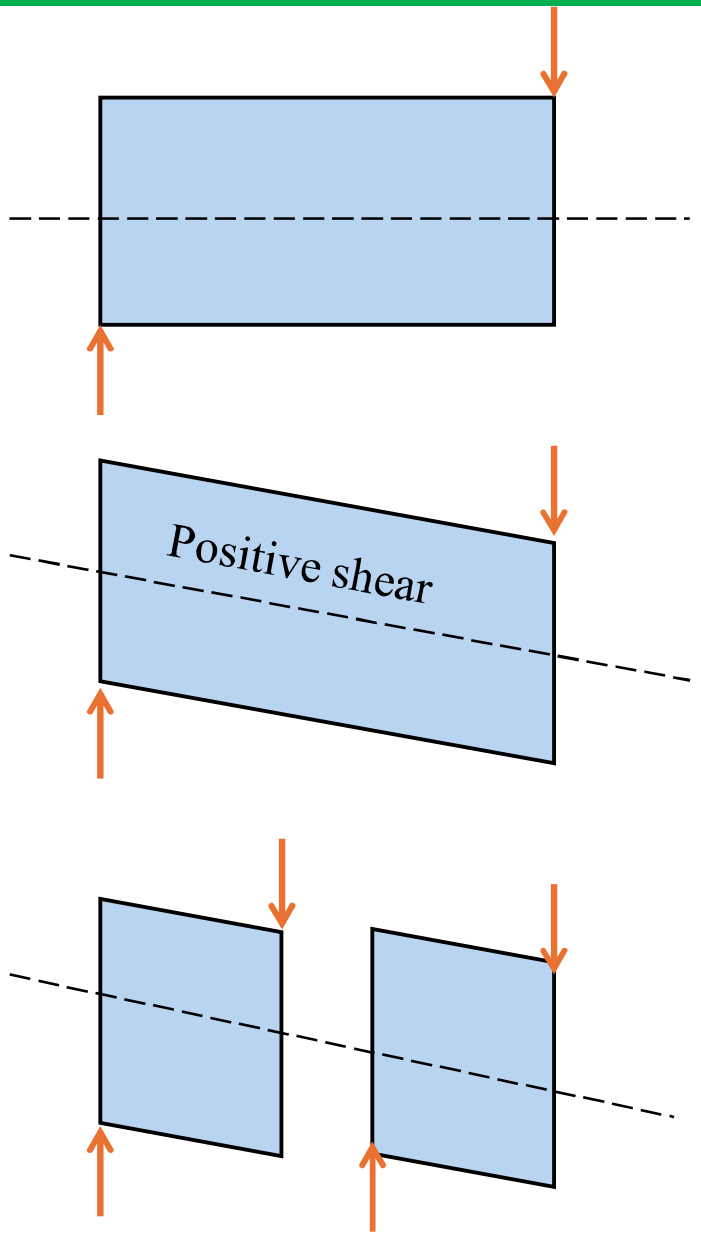


Bending

- Load: equal and opposite moments applied in planes containing the bar axis
- Deformation: relative rotation of adjacent cross sections around the axis perpendicular to bar axis
- Positive bending moment: sagging moment (upper half under compression)

Fundamental Action of External Force

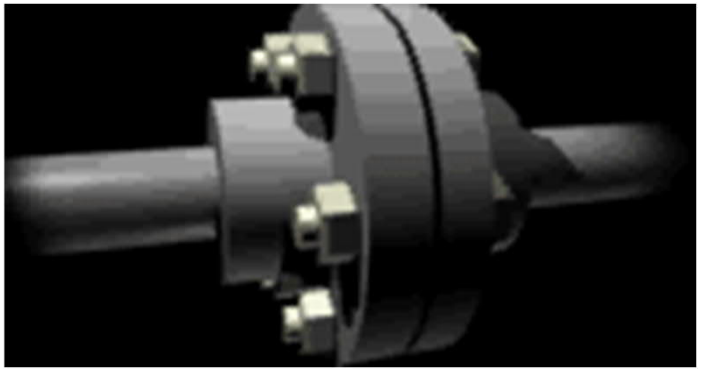


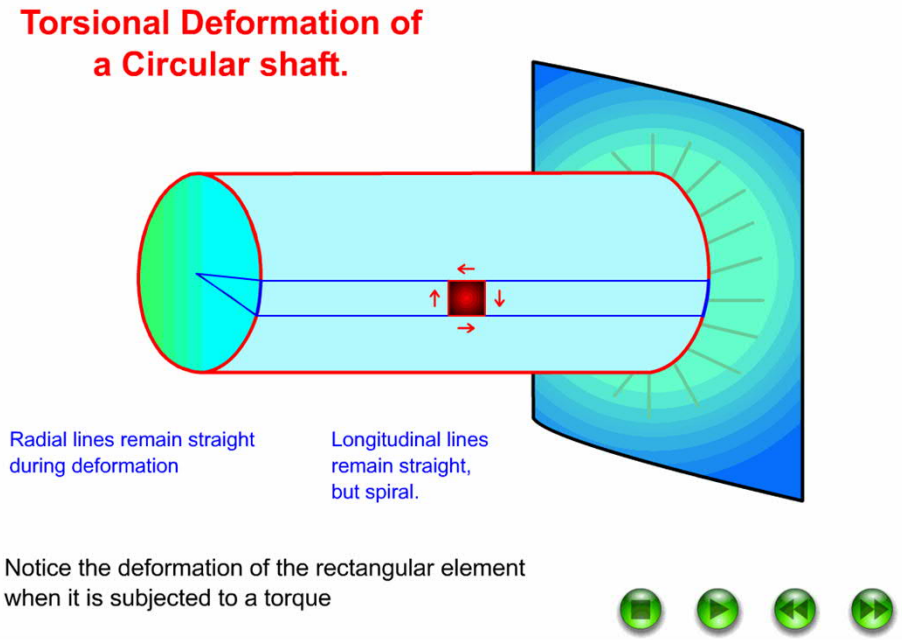


Shear

- Load: equal and opposite forces perpendicular to axis
- Deformation: relative shear of adjacent cross sections
- Positive shear: shear forces create clockwise moment

Fundamental Action of External Force





Torsion

- Load: equal and opposite moments applied as shown in the right figure.
- Deformation: relative rotation of adjacent cross sections around axis

Failure Criteria of Structures

- **Strength:** structural components should not fracture or yield under external loads
- **Stiffness:** maximum deformation should not exceed the allowable extent
- **Stability:** equilibrium state should be stable/ no buckling





Strength: Kobe earthquake (Japan), reinforced concrete column shear failure

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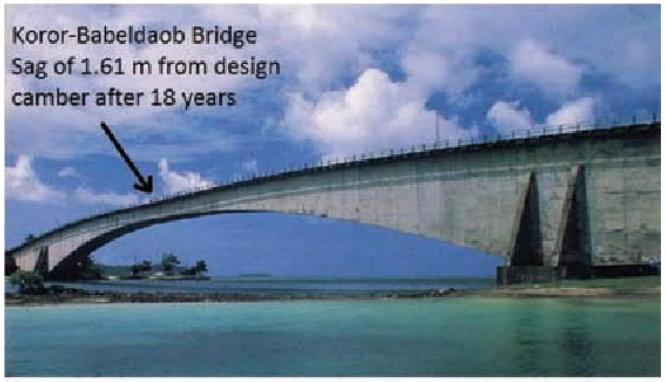


Tacoma bridge: torsional stiffness of section too low, wind-induced vibration cause fracture of cable



(Courtesy USGS archive)

California campus collapse in Northridge earthquake



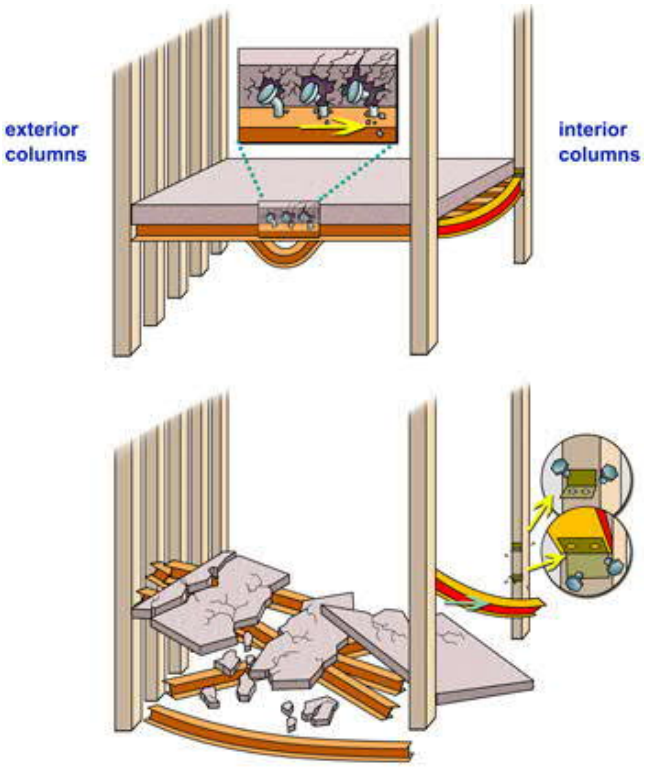


Koror-Babeldaob Bridge long-term deflection (1.6m) of prestressed concrete due to **creep**

Failure Criteria of Structures

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Thermal expansion led to the collapse of WTC Building 7. Heat from fires expanded steel beams on the east side of WTC 7, damaging the framing on multiple floors (top drawing). Eventually, a girder on Floor 13 lost its connection to a critical column and caused Floor 13 to collapse. The cascade of floor failures that followed left Column unsupported. When Column buckled, it initiated the global collapse of the building.

World Trade Center (WTC) collapse in 911

THE END